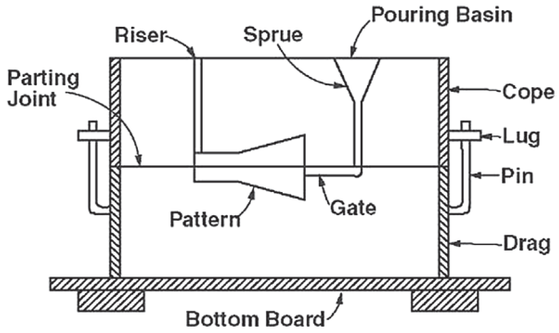


Green Sand Moulding

The following is the step by step procedure of making green sand mould using a split pattern.

1. One half of the pattern is placed on the moulding board.
2. Drag box is placed upside down on the board and parting sand is sprinkled over the pattern.
3. 20mm layer of facing sand is filled around the pattern. Then green sand is filled in the box.
4. Ramming is done uniformly by using rammer.
5. Excess sand is removed and levelled by strike off bar.
6. Vent holes are made. The box is tilted upside down.
7. The cope box and another half of the pattern are placed correctly. Parting sand is sprayed over the pattern.
8. The runner pin and riser pin are placed in the cope box at correct position. Then facing sand and moulding sand are filled.
9. Ramming is done uniformly. Vent holes are made.
10. Runner pin and riser pin are removed and pouring cup is made.
11. Cope and drag boxes are separated so as to remove the pattern.
12. Draw spike is driven into pattern pieces and shaken lightly in all direction. Then pattern pieces are withdrawn slowly.
13. Runner and gate are cut in drag portion.
14. Core is placed in the mould if necessary.
15. The cope and drag boxed are assembled in correct position and weight is placed over the cope. Now molten metal can be poured in the mould for producing casting.

Dry sand Moulding: Dry sand mould is obtained after heating the green sand mould. The procedure for making dry sand mould is same as that of green sand mould. The large moulds are heated by oxy-acetylene flame. Small moulds are heated in ovens.

Machine Moulding: Hand moulding is a slow process. It is suitable only for producing less number of casting. Ramming may not be done uniformly during moulding. Due to this, production time is increased. So machine moulding is best suitable for producing large number of similar castings. Accurate and better quality moulds can be prepared at faster rate by using machine moulding. The production cost is also reduced.





James Oliver
(August 28, 1823 – March 2, 1908) was an American inventor and industrialist best known for his creation of the South Bend Iron Work.

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